

Simulation Project 2

Mitas Stavros, Ashe Raymond-Barker

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Executive summary

This project used a SIMUL8 discrete-event simulation model to evaluate the performance of a laptop manufacturing system and test operational improvements. The baseline results showed that the system performed very poorly under the original policy. Around 19,545 orders arrived on average, but 17,744 were rejected, meaning that about 90.8% of demand could not be accepted. Only 1,807 orders were shipped, while average laptop-part completion time reached 3,194.41 minutes and completed laptops waited a further 2,406.46 minutes on average for order completion. The most serious problem was the case queue, with an average waiting time of 3,718.87 minutes, followed by the storage queue. These results showed that the system's poor performance was mainly caused by material shortages and inefficient flow logic rather than lack of total resource capacity.

The first improvement revised the aluminium inventory policy by increasing the reorder point from 10 to 30 and the reorder quantity from 50 to 70. This increased orders shipped from 1,807 to 2,481 and reduced the average case-queue waiting time by 63.2%, confirming that aluminium shortages were a major bottleneck. However, this also shifted congestion downstream, especially to the assembly stage.

The second improvement changed the storage-release logic so that stored items were prioritised when storage exceeded 15 items. This reduced average order completion time from 1,035.31 to 924.23 minutes and average laptop-part completion time from 2,023.04 to 1,831.82 minutes. It also reduced storage waiting time by about 35%, showing that the system became more stable and fairer.

The third improvement addressed the new downstream bottleneck by adding another workbench, increasing the aluminium reorder level to 90, and expanding the Power Assembly queue capacity. This reduced assembly-queue waiting by about 5%, cut average part completion time by around 15%, reduced maximum order completion time by 25%, and lowered variability by 8%. Overall, the results show that targeted changes to inventory policy, storage logic, and assembly capacity can substantially improve throughput, reduce delay, and make the system more balanced and reliable. Overall, the results show that the system's poor baseline performance was driven mainly by weak inventory control and inefficient queue logic rather than insufficient total resource capacity. The revised aluminium policy produced the largest throughput gains, while the revised storage policy further reduced delay and variability. Together, these changes demonstrate that targeted operational improvements can substantially improve system performance.

1 Data Analysis

1.1 Available Data

We were provided with 200 observations for each of the following processes in the computer assembly system:

- **Arrival Process:** Interarrival times in minutes for 200 orders.
- **Initial Phase:** Processing time in minutes for 200 items.
- **Placing the Keyboard and Mouse:** Processing time in minutes for 200 items.
- **Assembling the Case:** Processing time in minutes for 200 items.

The interarrival times were specified to follow an exponential distribution, while the distributions of the remaining processing phases were unknown. Therefore, in order to accurately simulate the computer assembly process, statistical analysis was carried out on the provided datasets to identify appropriate probability distributions that best represent the observed data.

1.2 Distribution Fitting Methodology

With only 200 data points available for each dataset, this limitation had to be considered when fitting probability distributions. For this reason we carried out Kolmogorov–Smirnov (KS) and Cramér–von Mises (CvM) tests alongside Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) tests for all candidate distributions.

The KS and CvM tests act as goodness-of-fit tests and determine whether a fitted distribution is consistent with the observed data. These provide a useful initial indication of whether a distribution is plausible. However, with a relatively small sample size of $n=200$, it is possible for multiple distributions to be considered acceptable. Therefore, AIC and BIC were used to distinguish between plausible fitted distributions. AIC was used as the primary model selection criterion, with BIC reported as a supporting metric.

An overview of the statistical tests used in the analysis is provided below. Detailed mathematical formulations of these tests can be found in Appendix A, whilst mathematical formulations of the distribution models we tested can be found in Appendix B.

Kolmogorov–Smirnov (KS) This test measures the largest difference between the cumulative distribution of the observed data and the cumulative distribution of the proposed theoretical model. If the difference is larger than some critical value, then the fitted distribution is rejected.

Cramér–von Mises (CvM) The Cramér–von Mises test measures the overall difference between the empirical and theoretical cumulative distributions. Where the KS test focuses on maximum difference, this test considers differences across the entire distribution, making it more sensitive to overall fit. Similarly, if the overall difference is larger than some critical value, then the fitted distribution is rejected.

Akaike Information Criterion (AIC) This model is used to compare competing statistical models by measuring the trade-off between model fit and model complexity. It does so by penalising models with too many parameters. A lower AIC value indicates a better trade-off between the two features.

Bayesian Information Criterion (BIC) The Bayesian Information Criterion (BIC) is similar to AIC but applies a stronger penalty for models with many parameters and therefore tends to favour simpler models. Similarly, a lower BIC value is preferred.

1.3 Arrival Process

To estimate the rate parameter of the exponential distribution, the sample mean of the interarrival times was calculated. This gave a value of

$$\bar{x} = 8.8340 \text{ minutes.}$$

For an exponential distribution, the mean is given by $E[X] = \frac{1}{\lambda}$, so the rate parameter can be estimated as

$$\hat{\lambda} = \frac{1}{\bar{x}} = 0.1132 \text{ per minute.}$$

Since it was specified that the arrival process follows an exponential distribution, model selection criteria such as AIC and BIC were not required. Instead, goodness-of-fit tests were used to determine whether the fitted parameter provides an appropriate representation of the data.

Distribution	KS Stat	KS p-value	CvM Stat	CvM p-value
Exponential	0.0447	0.8022	0.0443	0.9105

Table 1: Goodness-of-fit test results for arrival process

Table 1 shows that both the KS and CvM statistics are small and the corresponding p-values are large. This indicates that there is no evidence to reject the fitted exponential distribution with the estimated rate parameter.

The distribution overlay and Q-Q plot shown in Figure 1 further support that the fitted exponential distribution provides a good fit to the interarrival time data.

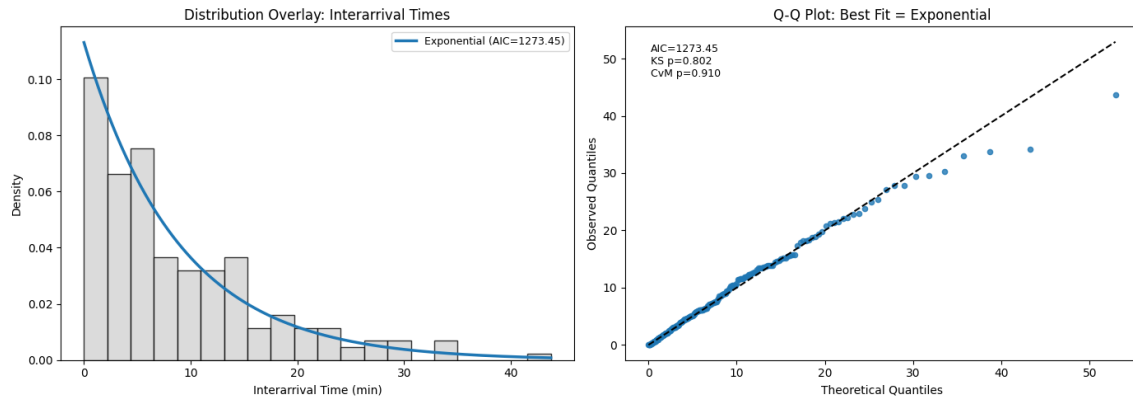


Figure 1: Distribution Overlay and Q-Q Plot of fitted exponential distribution

1.4 Initial Phase

We fitted four candidate distributions to the data: Weibull, Gamma, Log-Normal and Normal distributions. We note that there was a hard minimum duration in the dataset, with no observations falling below 1.24 minutes. As a result, a floating location parameter was allowed when fitting the positive-support distributions. This allows the distributions to be shifted so that they are not constrained to start at zero, which better reflects the observed data.

Results from the goodness-of-fit tests and model selection criteria for these fitted distributions are given in Table 2. The Weibull distribution provides lower KS and CvM statistics and higher p-values than the other candidate distributions. Furthermore, it has the lowest AIC and BIC values, indicating that it provides the best overall fit to the data.

The fitted parameters were $k = 1.8465$, $\lambda = 1.7093$, location = 1.2009. Therefore, the service times were modelled using a shifted Weibull distribution:

$$X = 1.20 + \text{Weibull}(k = 1.85, \lambda = 1.71).$$

Figure 2 further demonstrates that the shifted Weibull distribution provides a good representation of the observed data.

Distribution	KS Stat	KS p-value	CvM Stat	CvM p-value	AIC	BIC
Weibull	0.0350	0.9599	0.0517	0.8667	480.99	490.89
Gamma	0.0492	0.6999	0.0946	0.6127	483.59	493.48
Log-Normal	0.0497	0.6866	0.0858	0.6596	485.75	495.65
Normal	0.0506	0.6666	0.1614	0.3567	507.31	513.91

Table 2: Distribution fitting results for Initial Phase service times

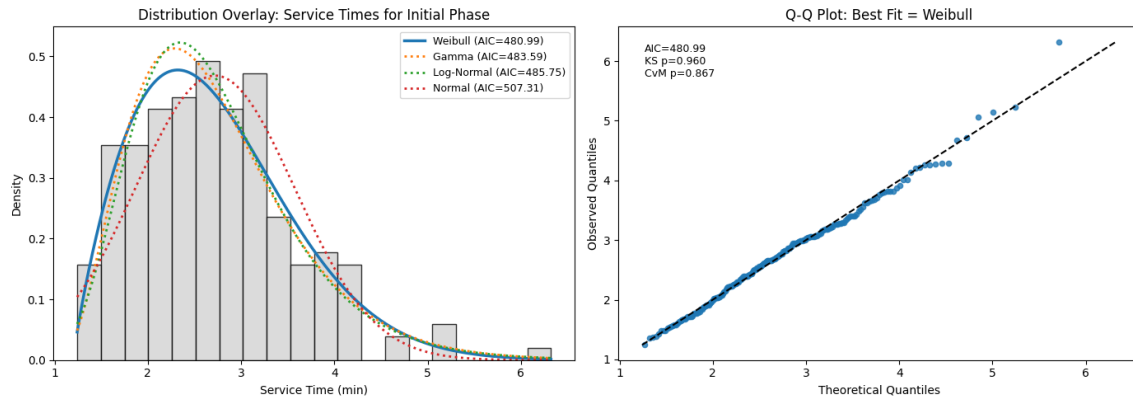


Figure 2: Distribution Overlay of fitted candidate distributions and Q-Q Plot of fitted Weibull distribution

1.5 Placing the Keyboard and Mouse

Here, Weibull, Gamma, Log-Normal and Normal distributions were again fitted to the data. Similarly to the Initial Phase dataset, there was a hard minimum duration in the data, with no observations falling below 4.04 minutes. Therefore, a floating location parameter was again allowed. However, this resulted in the Gamma and Log-Normal distributions producing negative location parameters of -20.4831 and -35.2916 respectively (see Appendix C, Table 18). These distributions were therefore considered unsuitable, as they would allow the model to generate negative service times, which are not physically meaningful in this context.

Comparison of the goodness-of-fit tests and model selection criteria in Table 3 shows that both the Normal and Weibull distributions provide a good fit to the data, with high p-values for the KS and CvM tests. However, the Normal distribution has slightly lower AIC and BIC values, indicating a better balance between model fit and model complexity. The service times were therefore modelled using the Normal distribution:

$$X \sim \mathcal{N}(7.8223, 1.5777^2).$$

Figure 3 further demonstrates that the Normal distribution provides a good fit to the observed data.

Distribution	KS Stat	KS p-value	CvM Stat	CvM p-value	AIC	BIC
Normal	0.0368	0.9402	0.0410	0.9283	753.97	760.57
Weibull	0.0359	0.9500	0.0398	0.9345	755.41	765.31
Gamma	0.0295	0.9929	0.0271	0.9849	755.61	765.50
Log-Normal	0.0297	0.9925	0.0273	0.9844	755.61	765.51

Table 3: Distribution fitting results for Placing Keyboard and Mouse service times

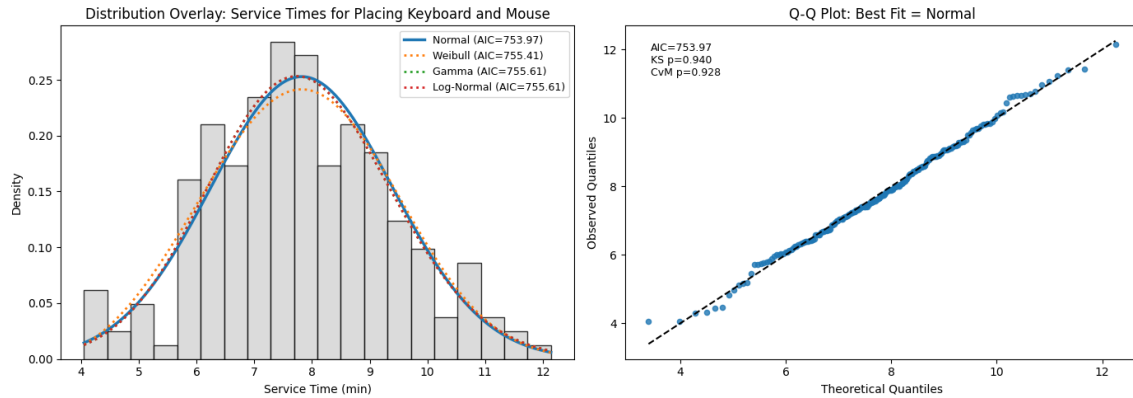


Figure 3: Distribution Overlay of fitted candidate distributions and Q-Q Plot of fitted Normal distribution

1.6 Assembling the Case

For this dataset, Exponential, Weibull, Gamma and Log-Normal distributions were fitted and tested. All candidate distributions except the Log-Normal distribution provided strong goodness-of-fit results, with high

p-values for both the KS and CvM tests, as shown in Table 4. Among the remaining candidate distributions, the Exponential distribution achieved the best balance between goodness-of-fit and model complexity, as indicated by the lowest AIC and BIC values.

We therefore chose to model this dataset using an Exponential distribution:

$$X \sim \text{Exponential}(\lambda = 0.2827).$$

Figure 4 further demonstrates that the exponential distribution provides a good fit to the observed data.

Distribution	KS Stat	KS p-value	CvM Stat	CvM p-value	AIC	BIC
Exponential	0.0383	0.9202	0.0278	0.9831	907.53	910.83
Gamma	0.0383	0.9195	0.0253	0.9891	909.08	915.67
Weibull	0.0396	0.8999	0.0257	0.9883	909.19	915.79
Log-Normal	0.0760	0.1886	0.3119	0.1252	940.06	946.65

Table 4: Distribution fitting results for Assembling the Case service times

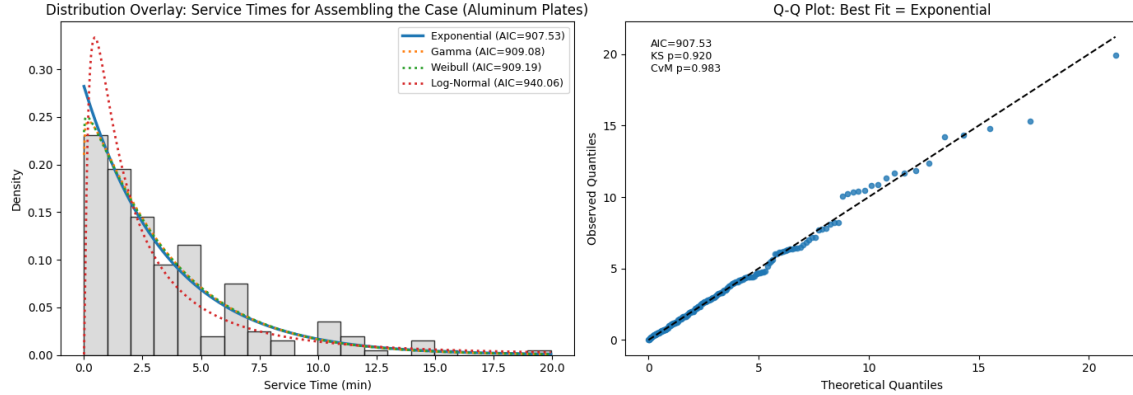


Figure 4: Distribution Overlay of fitted candidate distributions and Q-Q Plot of fitted exponential distribution

1.7 Summary of Selected Distributions

Table 5 details the choice of distribution and parameters selected for each process.

Process	Distribution	Parameters
Arrival Process	Exponential	$\lambda = 0.1132$
Initial Phase	Shifted Weibull	$k = 1.85$, $\lambda = 1.71$, location = 1.20
Keyboard and Mouse	Normal	$\mu = 7.8223$, $\sigma = 1.5777$
Assembling the Case	Exponential	$\lambda = 0.2827$

Table 5: Selected distributions for simulation input processes

Additional details of the data analysis process are provided in Appendix C. This includes tables containing the parameter estimates for all candidate distributions, Q-Q plots for the candidate distributions that were not selected, and cumulative distribution plots comparing the selected fitted distributions with the empirical data for each dataset.

2 Simulation Model Design

This section describes how the laptop manufacturing system was implemented in SIMUL8. The model was built in four main phases, following the flow of items from order arrival to final boxing. Each phase represents a major part of the production process and includes the routing, queueing, and control logic needed to reproduce the system described in the project brief.

2.1 Phase 1: Order Arrival and System Entry

The first phase of the model represents the arrival of customer orders and the decision of whether each order is accepted into the system. Order interarrival times follow an exponential distribution, while the size of each order is given by $X + 1$, where $X \sim \text{Binomial}(5, 0.6)$. Therefore, each order contains between 1 and 6 laptops. When an order arrives, the model generates its size and assigns a unique order ID using a global counter. The

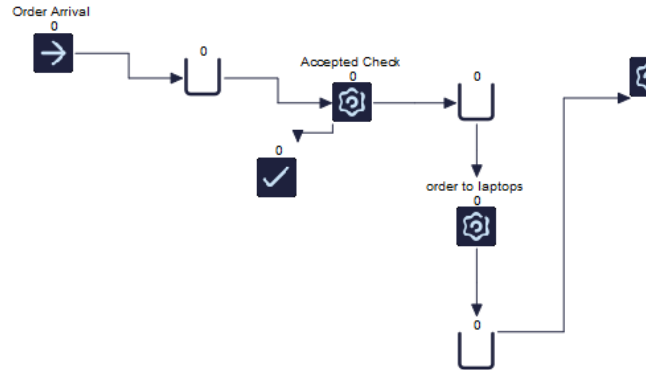


Figure 5: Phase 1

model then applies the company admission rule. If accepting the new order would cause the total number of laptops currently in the system to exceed 100, the order is rejected. Otherwise, the order is accepted and allowed to proceed. This rule limits the amount of work entering the system and directly affects the balance between accepted and rejected orders.

Each accepted order is then split into individual laptops. Each laptop becomes a separate work item and is assigned a unique laptop ID, while keeping the order ID of the order it belongs to. This is important because each laptop may have different specifications and must therefore be tracked individually throughout the production process. At the same time, the order ID is required so that all laptops belonging to the same order can later be identified for boxing.

Finally, each laptop is sent into two production branches that operate in parallel: the motherboard production branch and the case manufacturing branch. In this way, Phase 1 controls the entry of work into the system, creates the laptop-level entities used in the rest of the model, and starts the two main production branches.

2.2 Phase 2: Motherboard Production

The second phase of the model represents the motherboard production branch. This branch consists of three stages: *Initial Phase*, *Power Assembly*, and *Main Part Assembly*. The motherboard component of each laptop passes through these stages in sequence. In the *Initial Phase*, the motherboard is prepared for the later

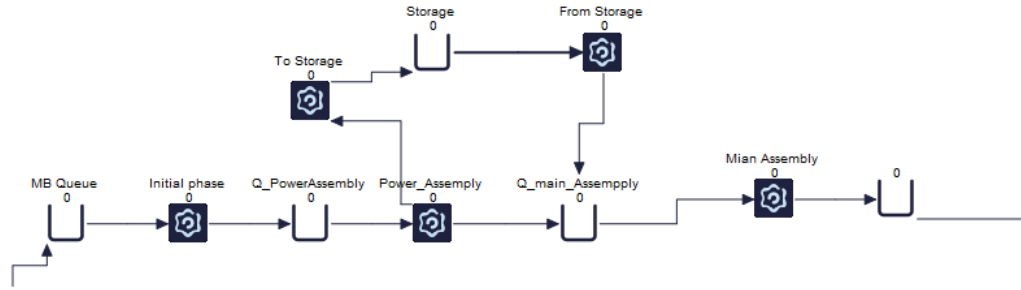


Figure 6: Phase 2

assembly operations. It then moves to *Power Assembly*, where the power-related components are added. A capacity restriction applies at this point: the queue in front of *Power Assembly* can hold at most 3 items. If this queue is full, the *Initial Phase* cannot release another motherboard and is blocked until space becomes available.

After *Power Assembly*, the motherboard proceeds to *Main Part Assembly*, where the main motherboard components are installed. The queue in front of this stage has a capacity of only 2 items. However, unlike the previous case, the upstream stage is not blocked when this queue is full. Instead, any additional motherboard is diverted to a storage area.

The storage area acts as an overflow buffer for items that cannot immediately enter *Main Part Assembly*. Sending an item to storage takes 2 minutes. Once an item is in storage, it remains there until the queue in front of *Main Part Assembly* becomes completely empty. Only then can items be moved back from storage into the queue. At that point, up to 2 stored items are returned, and each return move takes 3 minutes. Storage is therefore not used as a simple extension of the queue. It is a separate holding area from which items are released only when the main queue has been fully cleared.

This rule is important because it can generate substantial additional waiting in the motherboard branch. Items sent to storage do not return as soon as one space becomes available. Instead, they must wait until the queue is empty, which can delay their progress significantly.

Overall, Phase 2 models the motherboard branch as a sequential process with two important flow restrictions: blocking before *Power Assembly*, and overflow to storage before *Main Part Assembly*. These features are important in explaining the congestion and waiting times observed later in the results.

2.3 Phase 3: Case Manufacturing

The third phase of the model represents the case manufacturing branch. This phase begins with the production of the aluminum plates and then continues through two parallel branches: one for the upper part of the laptop case and one for the lower part. The first stage is *Manufacturing the Aluminum Plates*. This is

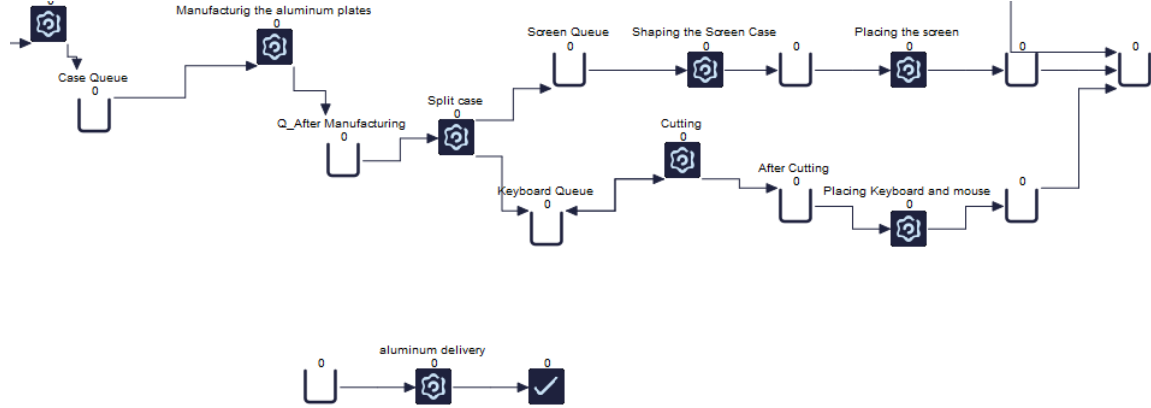


Figure 7: Phase 3

the starting point of case production for each accepted laptop. Since every laptop requires one complete case, and each case consists of an upper and a lower part, one aluminum block is used to begin the case manufacturing process for that laptop. This stage requires both the rolling machine and the cutting machine, and its processing time follows the specified deterministic value. It also consumes one aluminum block from inventory for each laptop entering the stage.

This stage is also linked to the aluminum inventory control policy. When the aluminum stock falls to 10 or fewer units, the model checks whether an aluminum order is already outstanding. If no order is currently outstanding, a new replenishment order is triggered and sent to the aluminum delivery process. This ensures that only one replenishment order is active at a time when the reorder level is reached.

After this stage, a zero-time dummy activity is used to split the laptop into two separate case components. One component represents the upper case and the other represents the lower case. These two components then move into two parallel branches.

The first branch represents the production of the upper part of the laptop case. The upper-case component moves first to *Shaping the Screen*, where the aluminum plate is shaped so that it can hold the screen. This activity uses the special machine and requires a type 3 worker. After shaping is completed, the component moves to *Placing the Screen*, where a type 1 worker fits the screen into the upper case. Once this step is complete, the upper part of the laptop case is ready for final assembly.

The second branch represents the production of the lower part of the laptop case. The lower-case component moves first to *Cutting*, where the aluminum plate is shaped so that the keyboard and mouse can later be installed. This activity also uses the special machine and requires a type 3 worker. A special feature of this branch is the rework logic. After the first cutting attempt, the keyholes may be too narrow with probability 0.2, in which case the component is sent back to the same queue for another attempt. If the component is processed a second time, the probability of rework falls to 0.1. After a third attempt, the component is always accepted. Therefore, no item can pass through this stage more than three times.

Once the cutting stage is completed successfully, the lower-case component moves to *Placing the Keyboard and Mouse*. In this stage, a type 1 worker fits the keyboard and mouse into the lower case. After this step, the lower part of the laptop case is also ready for final assembly.

Overall, Phase 3 models the case branch as one initial production stage followed by two parallel branches. The upper branch produces the screen side of the laptop, while the lower branch produces the keyboard side and includes the rework loop. Since both parts must be completed before the laptop can proceed to final assembly, delays in either branch can affect the overall flow of the system.

2.4 Phase 4: Main Assembly and Order Completion

The fourth phase of the model represents the final part of the system, where the completed components of each laptop are brought together, assembled, checked, and finally grouped back into customer orders for shipment. At the start of this phase, the model checks whether all three required components of the same

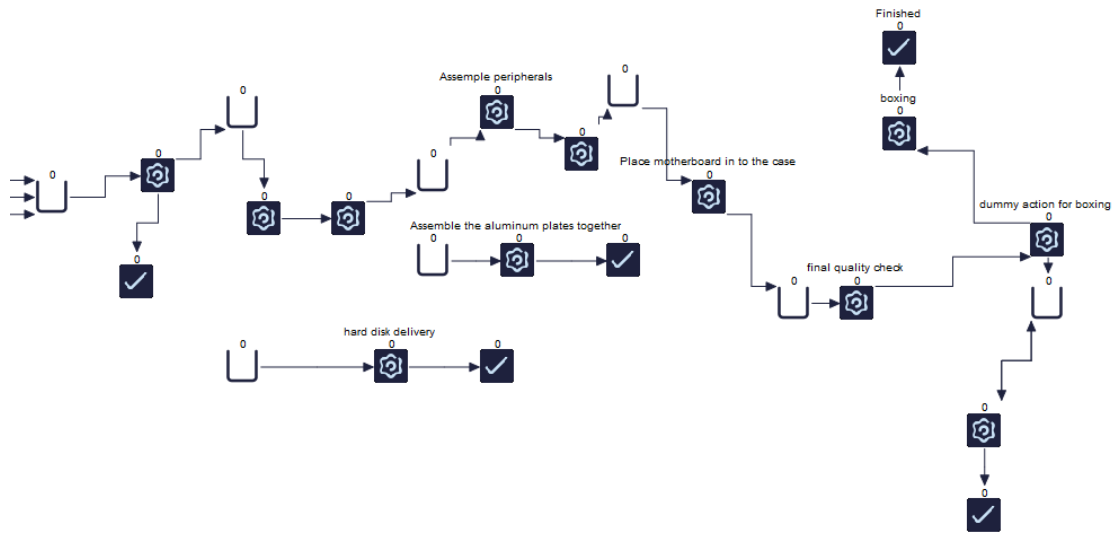


Figure 8: Phase 4

laptop have been completed. Each time one component finishes, the model updates the number of completed parts for that laptop ID. When the third and final component is completed, the three separate components continue as one laptop item. In this way, a laptop can only move forward when all of its required parts are ready.

Before the laptop enters the workbench section, the model uses zero-time dummy action to control its release. This does not represent a real processing step, but it is used to check whether the laptop is allowed to continue. First, it checks whether the required material is available. If the material is not available, the laptop is blocked from entering assembly. The same logic is also used to check whether a reorder of material is needed. The model then checks whether a workbench is free. Only when the required material is available and a workbench is empty can the laptop move into the assembly stage.

Once the laptop enters the workbench, the assembly stage is split into two parallel activities. In one activity, the hard disk and other peripherals are assembled. In the other activity, the aluminum plates are

assembled together. These two activities are performed in parallel while sharing the same workbench. The workbench remains occupied during this stage and is released only after the final assembly step is completed.

Before the laptop can move to *Place the Motherboard into the Case*, the model again uses dummy logic to check that both parallel assembly activities have finished. Only when both parts are ready does the laptop continue to this final assembly step. At this point, the motherboard is placed into the case and the laptop becomes fully assembled.

The assembled laptop then moves to the final quality check. After passing this stage, it enters a queue where the model checks whether all laptops belonging to the same order have been completed. This is done using the order ID. If the full order is not yet complete, the laptop waits until the remaining laptops from the same order are finished.

When all laptops belonging to the same order have been completed, the individual laptops are discarded and the model continues with one item representing the whole order. This order item then moves to the boxing stage, where it is packed and leaves the system.

Overall, Phase 4 combines laptop-level synchronization, workbench-based assembly, final quality control, and order-level completion logic. It is the stage that converts the outputs of the parallel production branches into completed and shipped customer orders.

3 Initial Output Analysis

3.1 Simulation Setup

The main objective of this project was to estimate the average cycle time of a laptop with a 95% confidence interval half-width of no greater than 1 minute, while also identifying changes capable of producing a meaningful improvement in overall system performance. To support this analysis, a range of Key Performance Indicators (KPIs) was recorded throughout the study, including cycle times, throughput measures, rejection rates, queue statistics, and resource utilisations. These measures were used both to establish the behaviour of the baseline system and to evaluate the effect of the proposed improvements.

An initial set of 10 independent replications was used to assess the variability of the model output and to determine whether the required precision target was practically attainable. For the selected cycle-time measure, the pilot results gave a mean of 613.00 minutes, with a 95% confidence interval of [598.77, 627.23], corresponding to a half-width of 14.23 minutes. The sample standard deviation across the 10 replications was 580.57 minutes.

Using the standard confidence interval expression

$$\bar{X} \pm t_{n-1,0.975} \frac{s}{\sqrt{n}},$$

the associated half-width is

$$h = t_{n-1,0.975} \frac{s}{\sqrt{n}}.$$

Rearranging for n gives

$$n = \left(\frac{t_{n-1,0.975}s}{h} \right)^2.$$

Substituting $s = 580.57$, $h = 1$, and $t_{9,0.975} = 2.262$ gives

$$n \approx \left(\frac{2.262 \times 580.57}{1} \right)^2 \approx 1.72 \times 10^6.$$

This indicates that approximately 1.7 million replications would be required to achieve the target half-width of 1 minute. Such a requirement is clearly not realistic within the scope of this project. The main reason is the substantial variability in the system output, which means that any meaningful reduction in confidence interval width can only be achieved through a very large increase in the number of replications. Although the 1-minute target provides a useful benchmark for the desired level of precision, it was not practically attainable here. Consequently, all scenarios were analysed using 10 independent replications, with the resulting averages and confidence intervals forming the basis of the comparisons reported later in the study.

The simulation run length was selected using the warm-up analysis shown in Figure 9. The plot indicates a pronounced transient period at the start of the simulation before the cycle-time measure settles into a more stable range. From this analysis, the warm-up period was estimated to be approximately 41,800 minutes, or about 29 days. This suggests that the model requires a relatively long initial period before its behaviour becomes representative of normal operation.

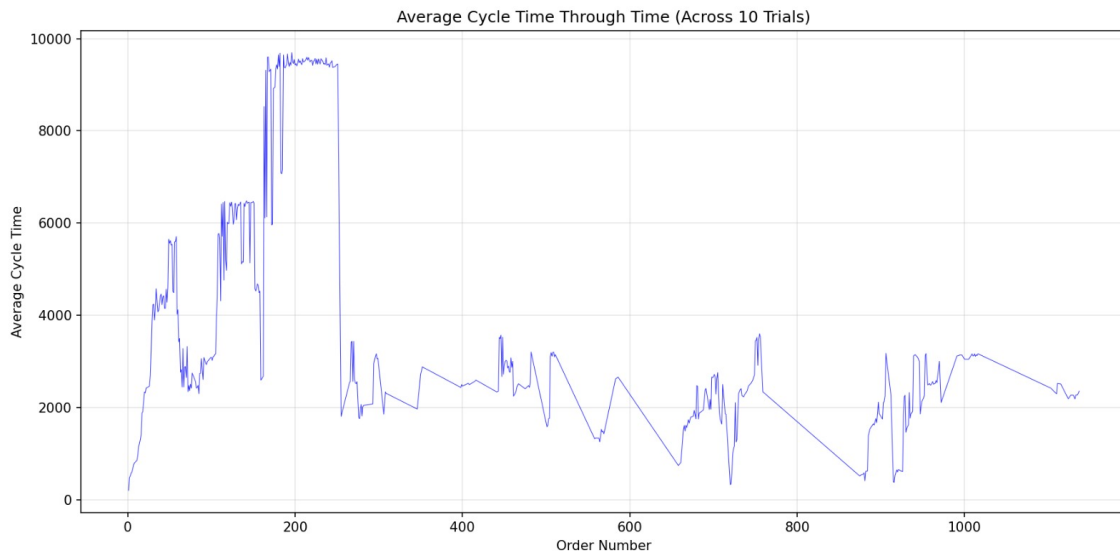


Figure 9: Warm-up analysis for the cycle-time measure across 10 trials. The plot suggests an initial transient period of approximately 41,800 minutes (about 29 days) before the output stabilises.

On this basis, a six-month simulation horizon was adopted for all model runs. This was considered sufficient because the estimated warm-up period accounts for only around 16% of the total run length, meaning that approximately 84% of each run reflects post-warm-up behaviour. In other words, the large majority of the simulation output is generated after the model has moved beyond its initial transient state. This was considered adequate for obtaining representative long-run measures of system performance.

A longer simulation horizon could also have been used, but it was not expected to materially affect the conclusions of the study. The model contains no seasonal effects or other slow-moving dynamics that would justify a substantially longer run length. Extending the horizon further would therefore mainly increase computational effort while offering only limited additional benefit in terms of output stability. A six-month horizon was therefore judged to provide an appropriate balance between representativeness and efficiency.

Overall, the adopted simulation setup reflects a practical compromise between statistical precision and

computational feasibility. Although the required half-width of 1 minute was not attainable without an unrealistically large number of replications, the use of 10 independent runs together with a six-month simulation horizon provided a sufficiently robust basis for analysing the baseline model and comparing the proposed improvement scenarios.

3.2 Baseline Model

The baseline model was run for six months, assuming 24-hour days and 5-day production weeks. In implementation, however, production was modelled as a continuous process over the full six-month horizon rather than stopping at the end of each working week and restarting at the beginning of the next. This baseline scenario provides the reference point against which later model refinements and operational changes are assessed.

Results for the baseline model are reported in Tables 6, 7, 8, 9, and 10. All reported confidence intervals were constructed from 10 independent simulation replications using the t-distribution:

$$CI = \bar{X} \pm t_{n-1, 0.975} \cdot \frac{s}{\sqrt{n}}.$$

Table 6 shows that approximately 90.8% of orders are rejected. This indicates that the 100-laptop work-in-progress (WIP) cap is active for most of the simulation, meaning that the system cannot process orders quickly enough to keep pace with demand. Since the acceptance rule depends on the amount of work already in the system, such a high rejection rate strongly suggests that excessive cycle times are a major contributor to poor throughput performance.

This is further supported by the cycle-time results in Table 7. At first sight, the average order completion time appears relatively low at 613 minutes, or around 10 hours. However, this value should be interpreted with caution. The completed-order statistic is driven mainly by orders that entered the system early in the simulation, before congestion had fully developed, and were therefore able to leave relatively quickly. As the run progresses, the system becomes increasingly congested and delays grow substantially. Many of the later orders, which are associated with much longer completion times, do not finish before the end of the simulation and are therefore excluded from the completed-order measure. As a result, the reported average order completion time understates the true severity of delay in the system.

This interpretation is consistent with the much larger values observed for the other cycle-time measures. The average completion time of a laptop part is 3,194.41 minutes, while the average time that a completed laptop waits for order completion is 2,406.46 minutes. Given the high arrival rate, such values are far too large to support acceptable throughput under the current WIP limit. The maximum observed times reinforce this conclusion. Although the mean order completion time is 613 minutes, the maximum reaches 12,200.72 minutes, while the corresponding maxima for part completion and completed-laptop waiting are 14,409.70 and 15,650.28 minutes respectively. Together with the large standard deviations, this indicates not only poor average performance, but also substantial variability, with some jobs experiencing extremely long delays.

The difference between the average completion time of a part and the average completion time of an order should also be interpreted carefully. Completed-order statistics only include orders that fully leave the system, whereas part-level statistics include individual items that may spend very long periods in the system even when the associated order is not completed before the simulation ends. For this reason, part-level times can exceed the corresponding order-level averages.

Metric	95% CI	Mean
Orders Arrived	[19,465, 19,626]	19,545
Orders Rejected	[17,662, 17,827]	17,744
Orders Boxed (shipped)	[1,793, 1,821]	1,807
Laptops through Quality Check	[7,012, 7,055]	7,034

Table 6: Throughput and rejection performance with 95% confidence intervals

Cycle	95% CI (mins)	Mean (mins)	Std Dev (mins)	Max Time (mins)
Completion of an Order	[598.77, 627.23]	613.00	580.57	12,200.72
Completion of a Part of Laptop	[2,863.76, 3,525.07]	3,194.41	2801.78	14,409.70
Completed laptop wait for order completion	[2,228.95, 2,583.98]	2,406.46	2717.07	15,650.28

Table 7: Cycle time results with 95% confidence intervals, standard deviations, and maximum observed times

In order to identify the sources of poor performance, it is useful to consider both resource utilisation and queue formation throughout the system. Table 8 shows that all workers and machines have very low utilisation rates. For example, Type 5 workers are utilised only 3.14% of the time on average, while the workbench is utilised only 12.23% of the time. This pattern suggests that the system is not primarily constrained by overall worker or machine capacity. Instead, jobs appear to be delayed upstream in particular parts of the process, preventing downstream resources from being used effectively.

At the same time, this combination of very high queueing delays (as shown in Table 9) and low utilisation is unusual. In a genuine capacity bottleneck, the constraining resources would normally be expected to show high utilisation. One possible explanation is that parts of the baseline model impose restricted flow through key stages, so that congestion is caused partly by process logic rather than by true resource scarcity. This means the baseline results should be interpreted cautiously: they are still useful for identifying where delays accumulate and which parts of the system are most sensitive, but they should not be treated as a perfect representation of real operational capacity.

Resource	Capacity	95% CI (%)	Utilization (%)	Avg Units Used
Type 1 Worker	7	[17.40, 17.52]	17.46	1.22
Type 2 Worker	3	[16.31, 16.63]	16.47	0.49
Type 3 Worker	3	[22.06, 22.20]	22.13	0.66
Type 4 Worker	2	[25.68, 25.88]	25.78	0.52
Type 5 Worker	2	[3.11, 3.16]	3.14	0.06
Roll & Cut Machine	2	[16.27, 16.38]	16.32	0.33
Special Machine	3	[22.06, 22.20]	22.13	0.66
Workbench	2	[12.19, 12.28]	12.23	0.24

Table 8: Resource utilisation with 95% confidence intervals

Table 9 identifies three main areas of concern: the case queue, the storage queue, and the motherboard (MB) queue. These queues arise from different mechanisms. The case queue is by far the most severe, with an average wait of 3,718.87 minutes and an average queue size of 151.61. This strongly suggests that the case line is the dominant bottleneck in the baseline system. A likely explanation is the availability of aluminium

inventory: without aluminium, case production cannot proceed, and the 15-hour replenishment delay can cause substantial accumulation in the queue.

Table 10 supports this theory, as it shows that around 140 aluminium deliveries are completed over the simulation horizon and that the delivery process is active for approximately 73% of the time. This implies that the current policies regarding aluminium supplies provide inadequate protection against stock depletion in the system. As a result, shortages in aluminium directly lead to long waits in the case manufacturing line.

The storage queue reflects a different problem. In the current logic, items are moved from storage to the main assembly queue only when that queue becomes empty. This creates an artificial delay for items already sent to storage, because newly completed items from power assembly may still join the main assembly queue directly whenever space is available. Consequently, stored items can remain delayed for extended periods, as reflected by the maximum storage wait of 3,286.56 minutes.

The MB queue is not the main bottleneck in the system, although it still shows noticeable delays. Compared to the case queue, its impact is much smaller. It is possible that part of this delay is caused by the way the model is structured, where only one process is allowed to occur at a time, rather than reflecting true production constraints. This is supported by the relatively low utilisation of Type 1 workers, suggesting that the motherboard line is not operating at full capacity. Despite this, the MB queue remains useful, as it highlights areas of the motherboard process that are sensitive to disruption.

Queue	Avg Wait (min)	Avg Queue Size	Max Queue Size	Max Wait (min)
After Cutting	1.82	0.07	3.40	25.02
Q_PowerAssembly	9.98	0.41	3.00	16.69
Q_main_Assembly	11.14	0.45	2.00	26.34
Queue for Peripherals	21.01	0.86	–	–
Keyboard Queue	64.05	4.93	27.50	179.68
MB Queue	408.32	16.61	616.10	2,466.55
Case Queue	3,718.87	151.61	638.80	15,624.95
Queue before assembly	207.74	8.49	–	–
Storage (overflow)	1,119.05	24.42	351.80	3,286.56

Table 9: Queue performance and bottleneck analysis ordered by process flow

Metric	95% CI	Mean
Aluminum deliveries completed	[139.90, 140.90]	140.40
Aluminum delivery Working (%)	[73.19, 73.55]	73.37

Table 10: Aluminum delivery performance with 95% confidence intervals

3.3 Diagnosis Summary

The baseline model reveals a number of operational weaknesses. The system rejects approximately 90.8% of arriving orders, while both order and part completion times are long and highly variable. These issues suggest that the production system is neither responsive enough nor stable enough to cope with the observed demand level.

Evidence gathered from output analysis of the baseline simulation suggests that the most urgent issues are the case queue and the storage logic. The case queue has an average waiting time of approximately 62

hours and is strongly linked to insufficient aluminium availability under the current replenishment policy. The storage logic creates an additional delay mechanism, with stored items waiting for the main assembly queue to empty before being released. Combined, these two problem create substantial congestion which propagates throughout the process and directly leads to poor throughput.

4 Improvements

4.1 Aluminium Supplies

The most urgent issue identified in the baseline model was the aluminium replenishment policy, which appeared to be a key driver of the excessive queue times observed in the Case Queue.

To address this, a revised inventory policy is proposed. The reorder point is increased from 10 to 30 aluminium blocks, and the reorder quantity is increased from 50 to 70. While this change would require additional storage capacity, and therefore incur some cost, it is expected to significantly reduce queueing delays. This, in turn, should increase throughput and overall system performance. Additionally, ordering larger quantities less frequently may reduce delivery-related costs, meaning that the additional storage costs are likely to be offset over time.

Metric	Baseline Mean	Improved Mean	Baseline 95% CI	Improved 95% CI	Change	% Change
Orders Arrived	19,545	19,545	[19,465, 19,626]	[19,465, 19,626]	0	0%
Orders Rejected	17,744	17,039	[17,662, 17,827]	[16,961, 17,117]	-705	-4.0%
Orders Boxed (shipped)	1,807	2,481	[1,793, 1,821]	[2,466, 2,496]	+674	+37.3%
Laptops through Quality Check	7,034	9,512	[7,012, 7,055]	[9,468, 9,556]	+2,478	+35.2%

Table 11: Comparison of baseline and improved throughput with 95% confidence intervals

Table 11 shows the impact of the revised aluminium policy on system throughput. There is a modest reduction of approximately 4% in rejected orders, and a substantial increase in both orders shipped and laptops passing through quality control, with improvements of 37.3% and 35.2% respectively. Importantly, the confidence intervals for the improved model do not overlap with those of the baseline model, indicating that these improvements are statistically significant and not due to random simulation variability.

Queue	Baseline Wait	Improved Wait	Change	Baseline Size	Improved Size	Change
Case Queue	3718.87	1367.45	-63.2%	151.61	76.59	-49.5%
Storage	1119.05	670.05	-40.1%	24.42	19.50	-20.1%
MB Queue	408.32	219.55	-46.2%	16.61	12.15	-26.8%
Assembly Queue (Workbench)	207.74	458.22	+120.6%	8.49	25.25	+197.4%
Keyboard Queue	64.05	88.29	+37.8%	4.93	9.20	+86.6%
Queue for Peripherals	21.01	22.04	+4.9%	0.86	1.21	+40.7%
Q_PowerAssembly	9.98	9.53	-4.5%	0.41	0.53	+29.3%
Q_main_Assembly	11.14	11.01	-1.2%	0.61	0.61	0%
After Cutting	1.82	1.93	+6.0%	0.07	0.11	+57.1%

Table 12: Comparison of queue performance between baseline and improved models

Table 12 demonstrates that this policy change significantly alleviates the primary bottleneck in the system. The Case Queue experiences substantial reductions in both average wait time (63.2%) and average queue size (49.5%), confirming that aluminium availability was a major constraint in the baseline model.

However, this improvement leads to increased pressure on downstream processes. In particular, the Assembly Queue experiences a significant increase in both wait time and queue size. This shift occurs because, once the case bottleneck is reduced, components flow more quickly through earlier stages of the system, resulting in more items arriving simultaneously at later stages. Similar increases are observed in the Keyboard Queue and other downstream queues, indicating that the system bottleneck has shifted rather than been eliminated.

Metric	Baseline Mean	Improved Mean	Baseline 95% CI	Improved 95% CI	Change
Aluminum deliveries completed	140.40	135.80	[139.90, 140.90]	[135.10, 136.50]	−3.3%
Aluminum delivery Working (%)	73.37	70.88	[73.19, 73.55]	[70.60, 71.10]	−2.49%

Table 13: Comparison of aluminum delivery performance with 95% confidence intervals

Table 13 presents results in line with our expectations. Both the number of deliveries and the amount of time spent waiting for an aluminium delivery reduces. Due to no overlapping of the new model’s confidence interval with the old one, we conclude that these are statistically significant changes. This suggests that the updated policy allows for more efficient use of delivery cycles while maintaining sufficient supply.

Interestingly, the average cycle time for completed orders increases from 613 minutes to 1,035 minutes. Although this appears counterintuitive, it can be explained by the increased throughput of the system. With more orders being successfully completed, the system now includes a greater proportion of longer-duration jobs that would previously not have been completed within the simulation period. As a result, the observed average increases.

In contrast, the average completion time for individual parts decreases significantly, from 3,194 minutes to 2,023 minutes (a reduction of approximately 37%), indicating that flow through the production process has improved substantially.

Overall, these results demonstrate that the revised aluminium replenishment policy leads to statistically significant improvements in system performance, particularly in reducing the primary bottleneck and increasing throughput, while also shifting congestion to later stages of the process.

4.2 Storage Logic

The second key issue identified was the queue logic controlling the storage system. In the baseline model, items are only transferred from storage to the main assembly queue when that queue becomes empty. This can lead to excessive waiting times for items placed in storage, as newly arriving items are prioritised over those already waiting. To address this, an additional routing constraint was introduced. When the number of items in storage exceeds 15, only items from storage are allowed to join the main assembly queue.

From a practical perspective, this change is expected to reduce the time that motherboards and partially completed laptops spend in storage, thereby improving flow consistency. It should also reduce variability in both part and order completion times, leading to a more predictable production system. Additionally, by limiting excessive accumulation in storage, we hoped this policy may reduce long-term storage requirements, potentially freeing capacity that could be repurposed for increased aluminium inventory.

Tables 14 and 15 demonstrate the positive effects of this change. Both the average cycle time of an order and for a part of a laptop to be completed decrease by a statistically significant amount, shown by the fact there is no overlapping of the 95% confidence intervals of the values. More importantly, there is a large decrease in the maximum cycle time and standard deviation of cycle time in both the order cycle and laptop

part completion cycles. This supports the interpretation that the revised storage logic improves fairness, as items are less likely to experience excessively long delays relative to others in the system.

Cycle	Imp 1 95% CI (min)	Imp 1 Mean	Imp 2 95% CI (min)	Imp 2 Mean	Change (min)	% Change
Completion of an Order	[994.42, 1076.19]	1035.31	[883.71, 964.75]	924.23	-111.08	-10.73%
Completion of a Part of a Laptop	[1874.25, 2171.83]	2023.04	[1727.10, 1936.55]	1831.82	-191.22	-9.45%
Completed laptop waiting for order completion	[906.10, 1063.84]	984.97	[914.67, 1049.48]	982.08	-2.89	-0.29%

Table 14: Comparison of cycle times between Improvement 1 and Improvement 2 with 95% confidence intervals

Cycle	Metric	Improvement 1	Improvement 2	Change
Completion of an Order	Max cycle time (min)	6,057	5,092.38	-15.9%
Completion of an Order	Cycle time std dev	316	263.96	-16.5%
Completion of a Part of a Laptop	Max cycle time (min)	14,409.70	13,239.37	-8.1%
Completion of a Part of a Laptop	Cycle time std dev	2801.78	2587.41	-7.7%

Table 15: Stability metrics for cycle times at End 8 and End 7 across improvements

In terms of the direct impact on the storage queue, Table 16 shows that while the maximum number of items stored remains largely unchanged, there is a substantial reduction in average waiting time of approximately 35%. The confidence intervals do not overlap, indicating that this reduction is statistically significant.

This suggests that the revised routing logic does not necessarily reduce peak storage requirements, but instead improves how efficiently storage is utilised. Items are processed more promptly, reducing unnecessary delays and preventing prolonged accumulation within the system. Overall, this reinforces the conclusion that the change improves flow consistency rather than simply reducing capacity requirements.

Metric	Imp 1 Mean	Imp 2 Mean	Imp 1 95% CI	Imp 2 95% CI	Change
Avg wait time (min)	670.05	433.76	[611.50, 728.60]	[400.58, 466.95]	-35.3%
Max number stored	241.10	240.10	[223.27, 258.93]	[215.78, 264.42]	0.005%

Table 16: Storage performance comparison between Improvement 1 and Improvement 2

4.3 Queues and Workbench Capacity

To address the shift of the bottleneck downstream, particularly at the assembly stage, we increased the number of workbenches. This allows more laptops to be assembled simultaneously and is expected to alleviate congestion at this stage. In addition, the aluminium reorder level was further increased to 90, and the capacity of the Power Assembly queue was expanded to three. These changes aim to improve overall system flow by reducing reliance on storage and ensuring that components can move more efficiently between stages.

The results show a modest reduction in the assembly queue bottleneck, with the average waiting time decreasing by approximately 5%. The relatively small improvement suggests that, while capacity has been increased, the assembly stage is not the sole limiting factor, and delays may still be influenced by upstream variability and synchronisation constraints between components.

Despite this, the combined changes lead to more substantial improvements in overall system performance. The average cycle time for computer part completion decreases by approximately 15%, while the maximum order completion time is reduced by 25%, from 5,092 minutes to 3,780 minutes. Additionally, the standard deviation decreases by 8%, indicating a reduction in variability. These results suggest that the system is not only operating more efficiently, but also more consistently, with fewer extreme delays.

Overall, these changes further improve system stability and fairness by smoothing the distribution of processing times and reducing variability. While the bottleneck is not entirely eliminated, it is mitigated, and the system becomes more balanced. This highlights that incremental capacity increases, when combined with improved flow and inventory policies, can lead to meaningful improvements in both throughput and reliability.

A Goodness-of-Fit Tests

A.1 Kolmogorov–Smirnov Test

Given a sample of size n with ordered observations $x_{(1)}, x_{(2)}, \dots, x_{(n)}$, the empirical cumulative distribution function is defined as

$$F_n(x) = \begin{cases} 0 & x < x_{(1)} \\ \frac{i}{n} & x_{(i)} \leq x < x_{(i+1)} \\ 1 & x \geq x_{(n)} \end{cases}$$

Then the maximum absolute difference between the empirical and theoretical cumulative distribution functions is given by:

$$D = \sup_x |F_n(x) - F(x)|$$

The null hypothesis is rejected if the test statistic exceeds a critical value

A.2 Cramér–von Mises Test

The test statistic is defined as

$$W^2 = n \int_{-\infty}^{\infty} (F_n(x) - F(x))^2 dF(x)$$

For ordered observations $x_{(1)}, x_{(2)}, \dots, x_{(n)}$, the statistic can be calculated using

$$W^2 = \frac{1}{12n} + \sum_{i=1}^n \left(F(x_{(i)}) - \frac{2i-1}{2n} \right)^2$$

The null hypothesis is rejected if the test statistic is larger than a critical value.

A.3 Akaike Information Criterion

It is defined as:

$$AIC = 2k - 2\ln(L)$$

where

- k is the number of parameters in the model,
- L is the maximum likelihood value of the model.

A.4 Bayesian Information Criterion

It is defined as:

$$BIC = k\ln(n) - 2\ln(L)$$

where

- k is the number of parameters,

- n is the number of observations,
- L is the maximum likelihood of the model.

The model with the lowest BIC value is preferred.

B Probability Distributions Used in the Simulation

B.1 Exponential Distribution

A random variable X follows an exponential distribution with rate parameter $\lambda > 0$ if its probability density function is

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

The mean and variance are given by

$$E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

If a location parameter ℓ is included, the distribution is shifted so that

$$X = \ell + Y, \quad Y \sim \text{Exponential}(\lambda),$$

B.2 Weibull Distribution

The Weibull distribution is often used to model service times and failure times. A random variable X follows a Weibull distribution with shape parameter $k > 0$ and scale parameter $\lambda > 0$ if its probability density function is

$$f(x) = \frac{k}{\lambda} \left(\frac{x}{\lambda} \right)^{k-1} e^{-(x/\lambda)^k}, \quad x \geq 0.$$

If a location parameter ℓ is included, the distribution becomes

$$X = \ell + Y, \quad Y \sim \text{Weibull}(k, \lambda),$$

B.3 Gamma Distribution

A random variable X follows a Gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\theta > 0$ if its probability density function is

$$f(x) = \frac{x^{\alpha-1} e^{-x/\theta}}{\Gamma(\alpha) \theta^\alpha}, \quad x \geq 0.$$

The mean and variance are

$$E[X] = \alpha\theta, \quad \text{Var}(X) = \alpha\theta^2.$$

If a location parameter ℓ is included, the distribution becomes

$$X = \ell + Y, \quad Y \sim \text{Gamma}(\alpha, \theta),$$

which shifts the distribution so that the support begins at $x = \ell$.

B.4 Normal Distribution

A random variable X follows a normal distribution with mean μ and standard deviation σ if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

If a location parameter ℓ is included, this shifts the mean of the distribution so that

$$X = \ell + Y, \quad Y \sim \mathcal{N}(\mu, \sigma^2).$$

B.5 Log-Normal Distribution

A random variable X follows a log-normal distribution if $\ln(X)$ is normally distributed. If $\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$, then the probability density function of X is

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0.$$

If a location parameter ℓ is included, the distribution becomes

$$X = \ell + Y, \quad Y \sim \text{LogNormal}(\mu, \sigma^2),$$

C Fitted Candidate Distribution Parameters and Additional Plots

C.1 Arrival Process

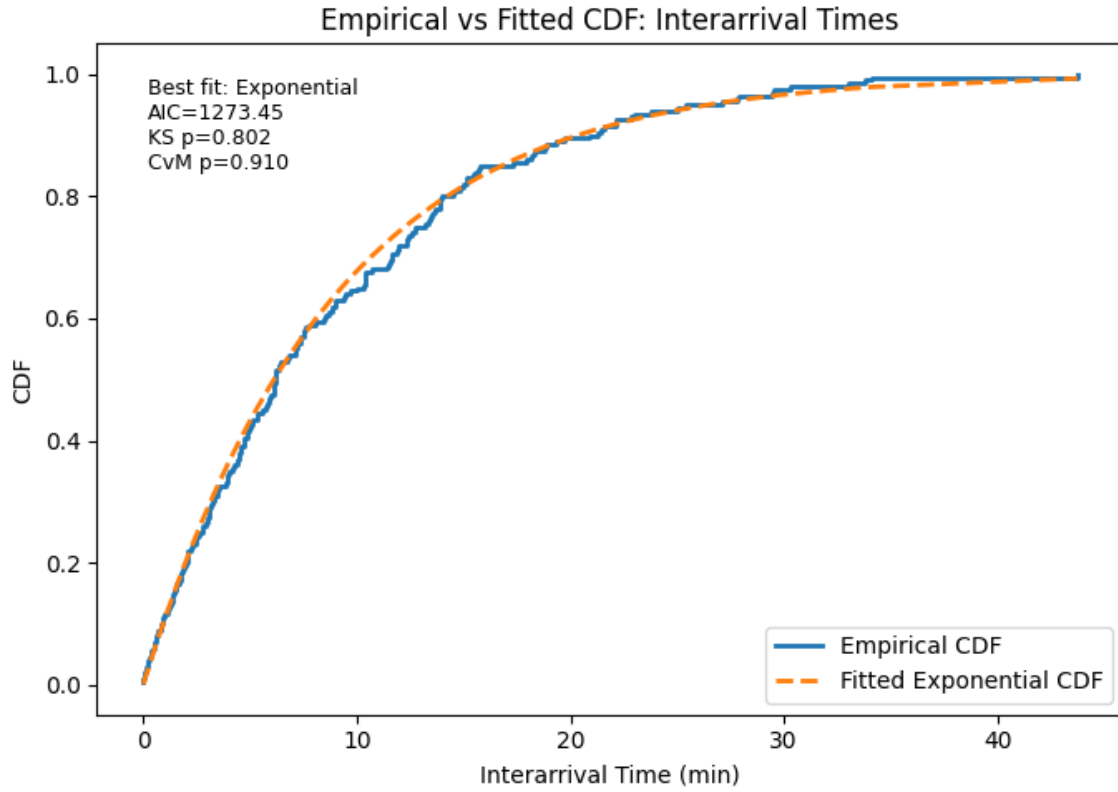


Figure 10: Q-Q Plots of unselected candidate fitted distributions

C.2 Initial Phase

Distribution	Parameters
Weibull	$k = 1.8465$, $\lambda = 1.7093$, location = 1.2009
Gamma	$\alpha = 4.0699$, $\theta = 0.4318$, location = 0.9626
Log-Normal	$\mu = \ln(2.5140)$, $\sigma = 0.3194$, location = 0.0756
Normal	$\mu = 2.7201$, $\sigma = 0.8516$

Table 17: Estimated distribution parameters for Initial Phase

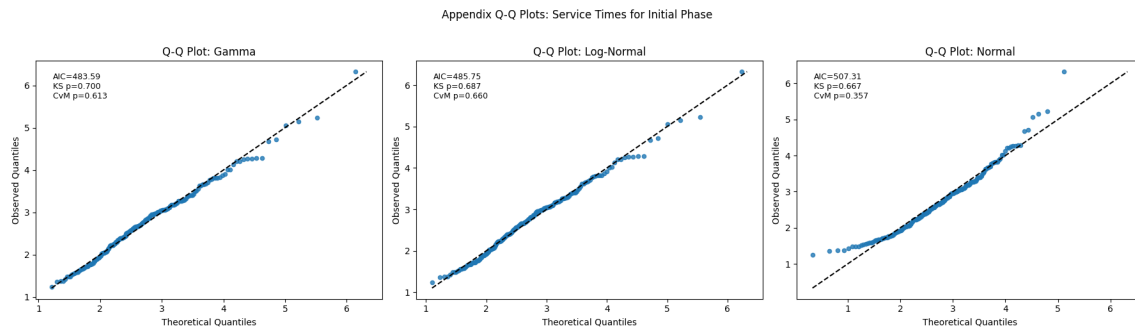


Figure 11: Q-Q Plots of unselected candidate fitted distributions

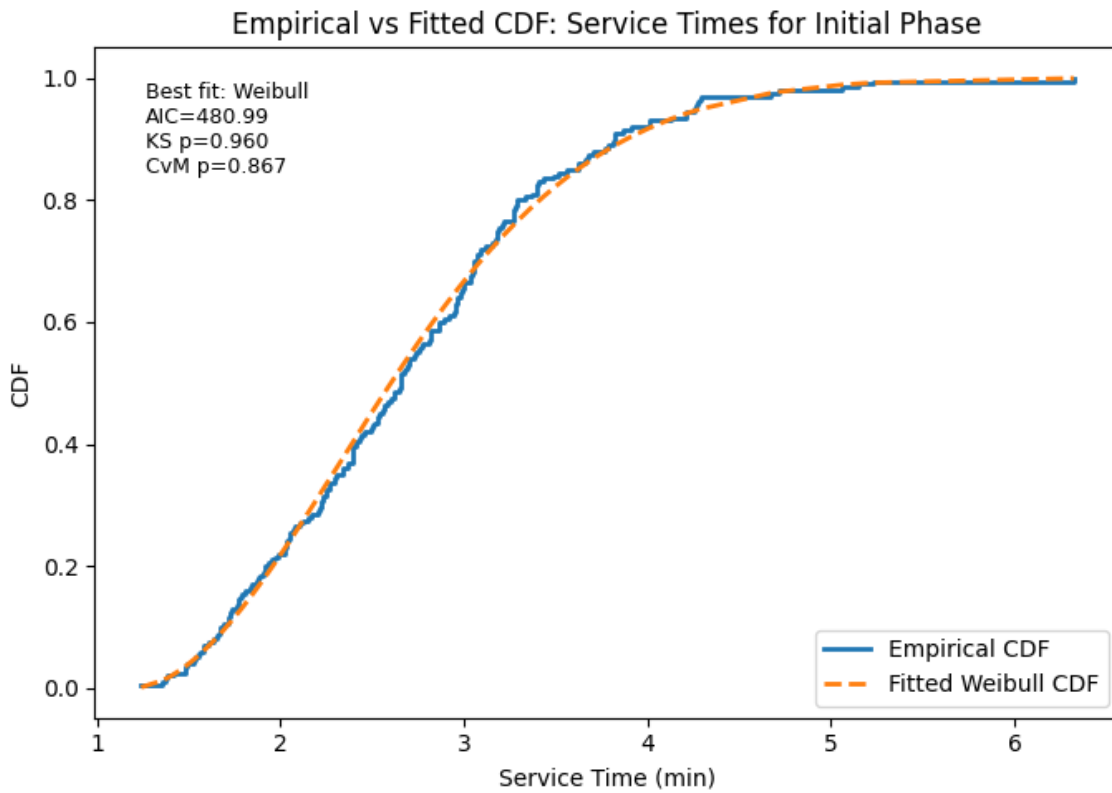


Figure 12: Cumulative Distribution graph comparison for the empirical and fitted CDF for initial phase service times

C.3 Placing the Keyboard and Mouse

Distribution	Parameters
Normal	$\mu = 7.8223, \sigma = 1.5777$
Weibull	$k = 3.3170, \lambda = 5.3174, \text{location} = 3.0488$
Gamma	$\alpha = 321.8249, \theta = 0.0880, \text{location} = -20.4831$
Log-Normal	$\mu = 3.763, \sigma = 0.0366, \text{location} = -35.2916$

Table 18: Estimated distribution parameters for Placing Keyboard and Mouse service times

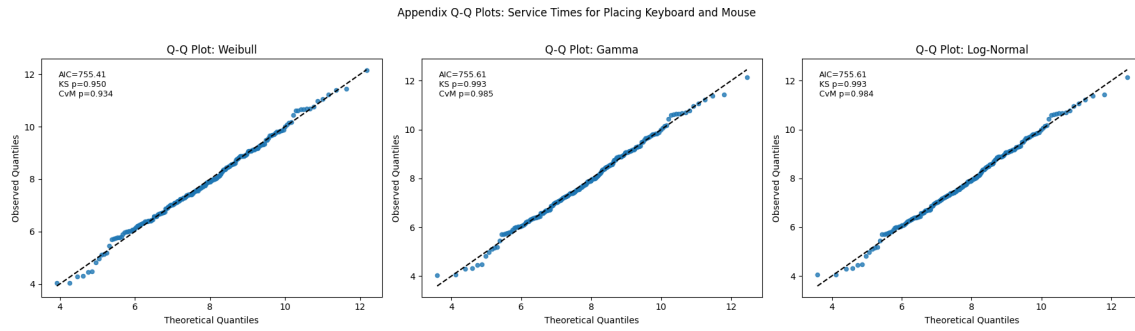


Figure 13: Q-Q Plots of unselected candidate fitted distributions

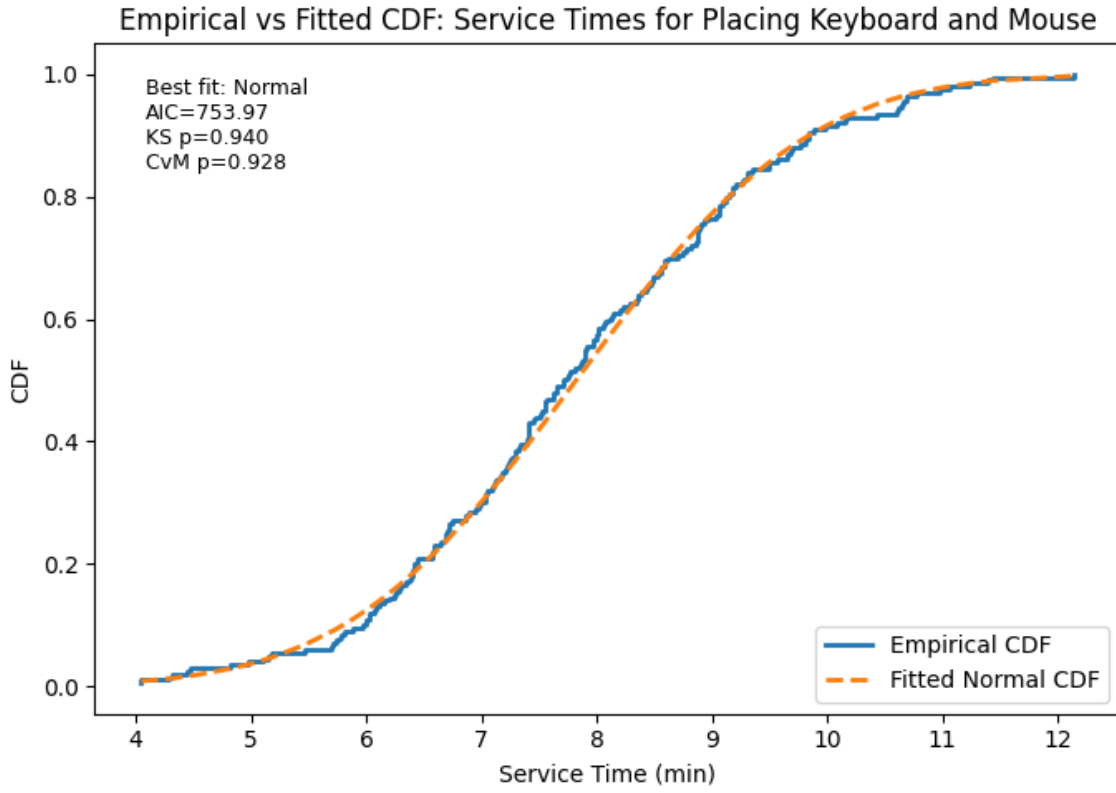


Figure 14: Cumulative Distribution graph comparison for the empirical and fitted CDF for keyboard and mouse placement service times

C.4 Assembling the Case

Distribution	Parameters
Exponential	$\lambda = 0.2827$
Gamma	$\alpha = 1.0620, \theta = 3.3324$
Weibull	$k = 1.0326, \lambda = 3.5860$
Log-Normal	$\mu = 0.7240, \sigma = 1.2179$

Table 19: Estimated distribution parameters for Assembling the Case service times

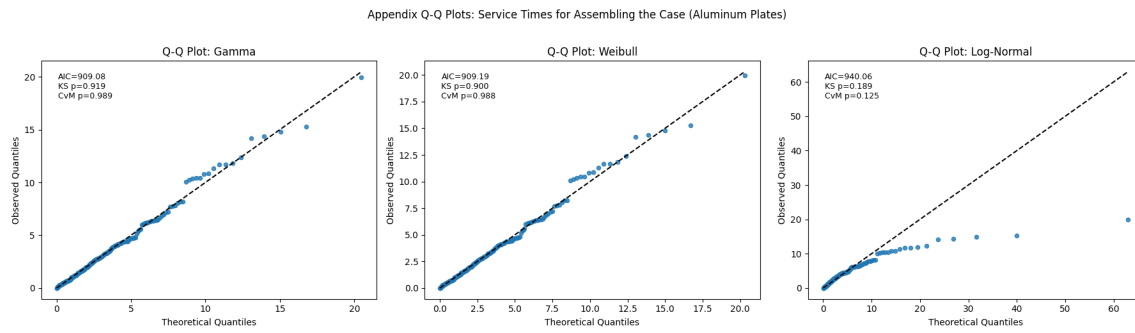


Figure 15: Q-Q Plots of unselected candidate fitted distributions

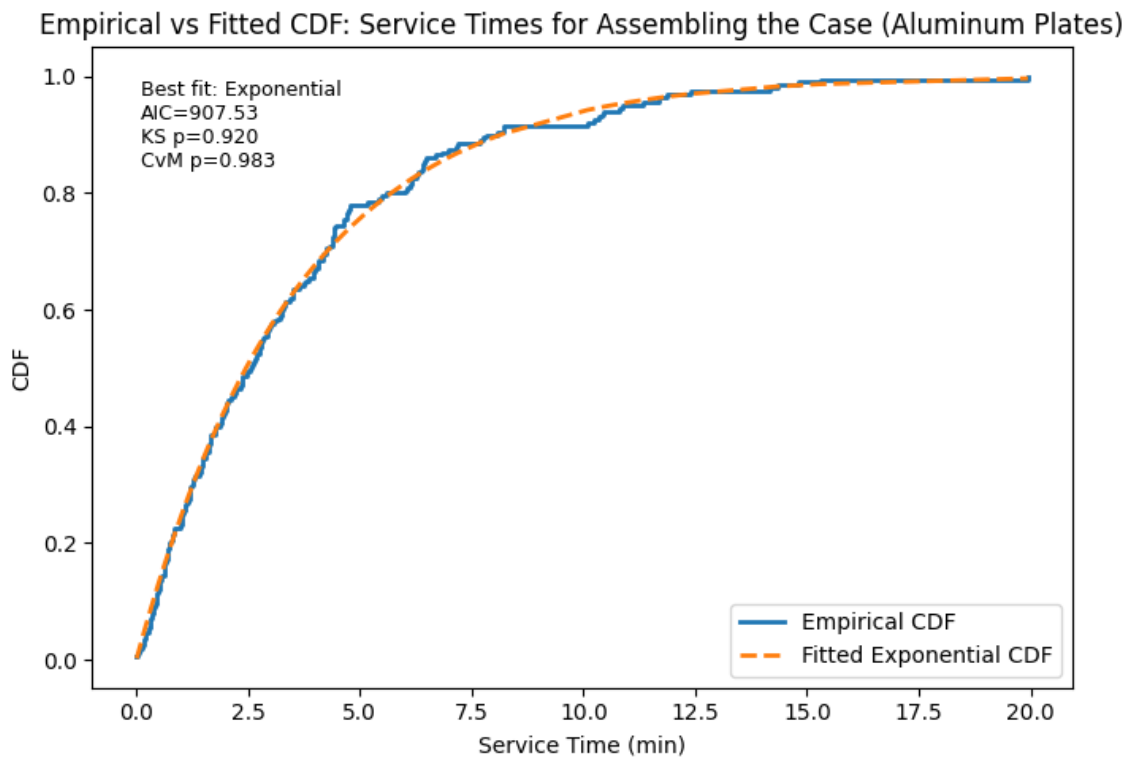


Figure 16: Cumulative Distribution graph comparison for the empirical and fitted CDF for assembling the case service times

D Full Simulation Model

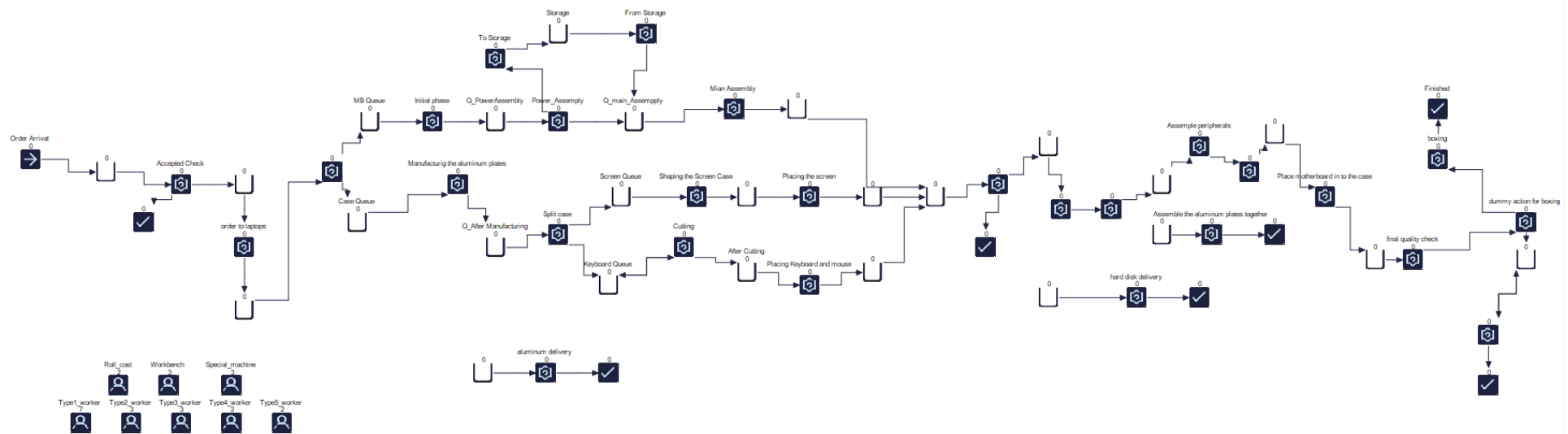


Figure 17: Full Simulation Model